Md. Fazle Rabbi Rahat

Marietta, GA | fazlerahat65.rf@gmail.com | Google Scholar↗| LinkedIn↗| Website↗

RESEARCH INTERESTS

Natural Language Processing · Large Language Models · Multimodal AI · AI Agents · AI Reliability

EDUCATION

- **Master of Science in Information Technology** **Jan 2026 - Expected Jul 2027**
  Kennesaw State University, Marietta, GA
  Proposed Thesis: *Agentic multimodal AI and retrieval-augmented decision support for hydroponic production systems*

- **Bachelor of Science in Computer Science & Engineering** **Feb 2020 - Aug 2025**
  Mawlana Bhashani Science and Technology University, Tangail, Bangladesh
  Thesis: *A Comparative Analysis of LoRA-Based Fine-Tuning Across Open-Source Language Models Under Resource Constraints for Medical QnA Automation*

RESEARCH EXPERIENCE

- **Graduate Research Assistant** **Jan 2026 - Present**
  iCPS Lab, Kennesaw State University, Advisor: Dr. Xu Tao
  - Developed an agentic multimodal framework that integrates visual observations, sensor data, and domain literature for real-time decision support in hydroponic production systems.
  - Designed a modular pipeline for multimodal context selection, hybrid retrieval, reranking, evidence aggregation, and tool-based sensor access.
  - Deployed lab-customized sensor infrastructure to support real-time monitoring and integration with the agentic decision-support framework.
  - Investigating efficiency issues on memory and context window, and reliability issues including hallucination, insufficient evidence, multimodal disagreement, and structured VLM consistency.
  - Currently fine-tuning language and vision-language model with domain-specific data.

- **Undergraduate Research Assistant** **Jan 2024 - July 2025**
  Mawlana Bhashani Science and Technology University, Advisor: Dr. Mostofa Kamal Nasir
  - Developed an LLM-based medical text interpretation system designed to assist patients in initial assessment
  - Fine-tuned small and large language models for medical-text interpretation under resource-constrained computing environments.
  - Investigated the effects of PEFT methods on model performance and their feasibility in comparison of a full fine-tuning.

- **Research Collaboration** **Oct 2023 - Nov 2025**
  Collaboration with a team in the *mHealth Lab* at Kennesaw State University
  - **IoMT Security Study:** Led the study and research-question formulation for a scoping review of ML approaches to IoMT security; developed the study taxonomy and led manuscript preparation and revisions. Resulted in a peer-reviewed paper at IEEE/ACM CHASE 2025.
  - **Project *DigiMindReady*:** Conducted a focused literature review on AI integration for military readiness and contributed to the proposed framework by designing and developing its LLM-based

features. Led manuscript preparation and subsequent revisions. This research resulted in a publication in *Military Medicine*.

- **Project *GoldenVaxShield*:** Conducted the systematic literature review on technology-enabled approaches to improving vaccination uptake and contributed to problem formulation and manuscript preparation. The research resulted in a publication in *Smart Health*.

PUBLICATIONS

- **Rahat, M. F. R.**, Hasan, M. M., Xie, Z., & Sakib, N. (2025). Optimizing IoMT Security Using Machine Learning: A Use Case-Based Scoping Review toward Smart and Connected Health. ACM/IEEE International Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), 371–376. https://doi.org/10.1145/3721201.3724412 [Conference paper↗]
  - Presented our paper in the conference.
- Hasan, M. M., **Rahat, M. F. R.**, Anjum, N., Sakib, N., & Moore, B. A. (2026). DigiMindReady: Enhancing military readiness through Edge AI-Driven wellness, education, and digital discipline through Privacy-First mHealth Innovation. Military Medicine, 191(9–10), e2039–e2048. https://doi.org/10.1093/milmed/usag101 [Journal Article↗]
- Hasan, M. M., **Rahat, M. F. R.**, Anjum, N., Ramos, M. D., & Sakib, N. (2026). Enhancing vaccination uptake for healthy aging through context-aware mHealth systems: A systematic literature review for future smart healthcare. *Smart Health*, 41, 100665. https://doi.org/10.1016/j.smhl.2026.100665 [Journal Article↗]

TEACHING EXPERIENCE

- **Graduate Teaching Assistant** **Jan 2026 - Present**
  Department of Information Technology, Kennesaw State University
  Courses: Advanced Application Development (Summer 2026), Ethical Hacking Defense (Fall 2026), OS Concepts & Administration (Fall 2026)
  Grading programming assignments and projects with detailed technical feedback, assisting with instructional materials, and support students through office hours.

TECHNICAL SKILLS

- **AI / ML:** LLMs, VLMs, Embeddings, Vector Databases, RAG, Agentic Systems
- **Tools:** PyTorch, Hugging Face Transformers, Ollama, LangChain
- **Programming:** Python, JavaScript, Java, C++
- **Application Development:** React, Flutter, SwiftUI